

What This Project Actually Asked

A plain-language summary of the current satellite-image segmentation study

Research question:

Can simple artificial changes to satellite training images, or showing the model more examples of a rare land type, improve a fixed model?

In everyday language

We trained a computer program to look at a satellite image and label each pixel as a land type, such as building, road, water, forest, agriculture, or barren ground. Then we asked whether changing some training images in simple ways would make that program better.

The original, broader idea was about whether synthetic images help the model work in unfamiliar places or conditions. This first study tested only the **conditions** part: brightness changes, image rotations, and a method that shows more rare Barren-land examples. It did not yet test a completely different geographic location.

What we used

Item	What it means
Training data	1,366 labeled Rural LoveDA satellite images
Validation data	992 separate Rural images used only to check performance
Model	A compact U-Net: a standard image-segmentation model
Fair comparison	Same model, image size, training length, and training budget for each experiment

What we tested

Test	Change made	Why we tested it
Brightness	10% of training images were slightly darker or brighter copies.	To see if the model handled different lighting better.
Rotation	10% of training images were rotated by 90, 180, or 270 degrees.	To see if the model became less sensitive to orientation.
Barren-focused sampling	Images containing rare Barren land were shown more often during training.	To help the model learn a land type it was missing.

What we found

Result	Simple meaning
Brightness did not help overall	Across three repeated runs, overall quality fell slightly: 24.64% to 23.71% mean mIoU. It also did not show a dependable improvement on brighter or darker validation images.
Rotation made results worse	In the first rotation test, overall quality dropped from 24.42% to 21.54% mIoU. We stopped instead of spending time on more runs of an unpromising idea.
More Barren examples helped Barren	The normal model did not recognize Barren land at all. The focused sampling raised Barren recognition to 3.12% IoU on average across three runs.
But there was a tradeoff	Focused Barren sampling kept overall mIoU nearly the same (24.64% to 24.60%) but lowered simple pixel accuracy by 3.97 points. It helps a rare class, not every result at once.

Bottom line

The answer for this project is: Simple brightness and rotation copies were not useful for this model. Showing it more examples of rare Barren land helped it recognize that rare land type, but came with a small cost to general accuracy. In short, targeted help for a rare class worked better than generic image changes.

What this does not answer yet

This work does **not** yet prove whether synthetic imagery helps in a totally different city, country, or geographic region. Every experiment here trained and validated on the Rural LoveDA split. A future study would train on Rural images and test on a genuinely different domain, such as Urban LoveDA or another satellite dataset.

Metric note: mIoU is a balanced score for how well the model outlines each land type. Pixel accuracy is the percentage of individual pixels labeled correctly. Higher is better for both.